Civil Engineering Level 6

Calculus and Differential Methods

Assignment 1 - Questions Only

Instructions: Answer all questions. Show all working clearly. Use proper mathematical notation.

Section A: Short Questions

1. State two uses of differentiation in civil engineering calculations. **(4 marks)**
2. Differentiate: **(4 marks)**

$$y = 6x^3 - 5x^2 + 9x - 4$$

3. Differentiate: **(4 marks)**

$$y = (3x + 2)^5$$

4. Find the gradient of the curve at x = 4: **(4 marks)**

$$y = 2x^2 + 3x + 1$$

5. Integrate with respect to x: **(4 marks)**

$$8x^3 - 6x + 2$$

6. Evaluate: **(4 marks)**

$$\int_0^3 (2x + 5)\, dx$$

7. Define an ordinary differential equation. **(4 marks)**
8. State the standard form of a first-order linear differential equation. **(4 marks)**

$$\frac{dy}{dx} + P(x)y = Q(x)$$

9. Write the Newton-Raphson formula. **(4 marks)**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

10. State one condition required when applying Simpson's one-third rule. **(4 marks)**

Section B: Structured Civil Engineering Applications

11. A proposed access road has a vertical profile approximated by:

$$y = 0.04x^2 - 1.2x + 115$$

where x is horizontal distance in metres and y is level in metres. a) Find the gradient function. b) Determine the gradient at x = 10 m. c) Find the chainage where the profile has a stationary level. d) Interpret the result for road design. **(15 marks)**

12. A drainage channel has a cross-section modelled by:

$$y = 0.5x^2 + 1, \quad 0 \le x \le 4$$

a) Sketch the general shape of the curve. b) Use integration to determine the area under the curve. c) Explain how this area may relate to construction measurements. **(15 marks)**

13. The rate of settlement of a soil layer is modelled by:

$$\frac{ds}{dt} = 0.6t$$

where s is settlement in mm and t is time in months. a) Find s in terms of t. b) If settlement is zero at t = 0, determine the constant of integration. c) Calculate settlement after 5 months. d) State one limitation of this model. **(15 marks)**

14. The function below has a root between x = 2 and x = 3:

$$f(x) = x^3 - 4x - 9$$

a) Show that the root lies in this interval. b) Use one iteration of the bisection method. c) Use one Newton-Raphson iteration starting at x = 2.5. d) Explain why numerical methods are useful in engineering. **(15 marks)**